

Arya Yae

Ph.D. in Mathematics

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Research Interests

Gauge Theory, Moduli Spaces, Algebraic Geometry, Geometric Invariant Theory

Education

- 2019–2026 **Ph.D. in Mathematics**, *University of Oregon*, *GPA: 4.00*
Dissertation: “Hyperkähler Degeneration from Parabolic $SL(2, \mathbb{C})$ -Higgs Bundle Moduli Spaces to Hyperpolygon Spaces.” Advisor: Dr. Laura Fredrickson
- 2019 **B.S. in Mathematics**, *Seattle University*, *summa cum laude*, *GPA: 4.00*
Undergraduate thesis: “Set Theory and Mathematical Logic.” Advisor: Dr. Dylan Helliwell
- 2017 **Associate of Arts**, *Bellevue College*, *GPA: 3.96*

Publications

- 2026 **Fredrickson, Yae.** Star-Shaped Nakajima Quiver Varieties, Parabolic Higgs Bundle Moduli Spaces, and their Holomorphic Symplectic Structures (submitted). [arXiv:2606.23369](https://arxiv.org/abs/2606.23369)
- 2026 **Fredrickson, Yae.** Hyperkähler Degeneration from Parabolic $SL(2, \mathbb{C})$ -Higgs Bundle Moduli Spaces on the Punctured Sphere to Hyperpolygon Spaces. *Communications in Mathematical Physics* (accepted and appearing soon). [arXiv:2601.10656](https://arxiv.org/abs/2601.10656)
- 2022 **Ding, Helliwell, Pandya, Yae.** Further Study on Domains and Quasihyperbolic Distances. *J Inequal Appl* 2022, 140 (2022). [arXiv:2207.08039](https://arxiv.org/abs/2207.08039)
- In prep. **Huchala, Yae.** Pole Order Reduction via Griffiths Residues on Grassmannians

Fellowships and Awards

- 2021, 2023–2025 NSF Research and Training Grant (RTG) Fellowship (University of Oregon), summers
- 2024 Anderson Teaching Award (University of Oregon)
- 2023 Anderson Research Award (University of Oregon)
- 2020 Civin Award (University of Oregon)
- 2019 Mirbagheri-Yandl Achievement Award (Seattle University)
- 2017 Lebon Prize for Excellence in Mathematics (Bellevue College)

Talks

Invited

- Jun 2026 Kavli Institute for the Physics and Mathematics of the Universe (conference): *Star-Shaped Nakajima Quiver Varieties and Parabolic Hitchin Moduli Spaces*
- Jun 2026 Osaka Metropolitan University (conference): *Metric Degeneration from Higgs Bundle Moduli Spaces to Hyperpolygon Spaces*
- Mar 2026 University of York Geometry/Topology Seminar

Dec 2025 University of Massachusetts Amherst: *Metric Degeneration in Moduli Spaces of Parabolic Higgs Bundles*

Nov 2025 University of Arizona Probability and Mathematical Physics Seminar

Oct 2025 AMS Special Session on Gauge Theory and Low-Dimensional Topology: *Hyperkähler Degeneration of ALG- D_4 Gravitational Instantons*

Oct 2025 Oregon State University: *Some Constructions of Hyperkähler 4-Manifolds*
 Departmental (University of Oregon)

Feb 2026 *Metric Degeneration from Higgs Bundle Moduli Spaces to Hyperpolygon Spaces*

Oct 2025 *Uniformization Metrics and Kähler-Einstein Connections*

May 2025 *Metric Degeneration from ALG to ALE Gravitational Instantons*

Apr 2025 Student Teaching Seminar: *Assessment Practices*

Feb 2025 *What is a Moduli Space?*

Jan 2025 *Metrics on Moduli Spaces via Symplectic Reduction*

Teaching

Mentor for Directed Reading Programs

Ricci Curvature Flow and Geodesics, with 2 students — simulating singular Ricci flow in Python (2026); *Axiomatic Set Theory* (2023)

Instructor of Record

<i>Intro to Proof</i>	Winter 2026, Spring 2026
<i>Discrete Math I</i>	Winter 2025, Spring 2025, Fall 2026
<i>Calculus III: Sequences and Series</i>	Fall 2024
<i>Intro to Linear Algebra</i>	Fall 2023
<i>Several-Variable Calculus I</i>	Summer 2022
<i>Calculus II: Integral Calculus</i>	Winter 2022, Spring 2022
<i>Business Calculus II</i>	Spring 2021
<i>College Algebra</i>	Winter 2020, Spring 2020, Winter 2021
<i>Precalculus: Elementary Functions</i>	Fall 2020

Teaching Assistant

Business Calculus I (Spring 2024), *Calculus I: Differential Calculus* (Fall 2021), *College Algebra* (Fall 2019)

Grader

Intro to Topology II (Winter 2023), *Statistical Methods I* (Winter 2023), *Intro to Topology I* (Fall 2022), *Fundamentals of Analysis I* (Fall 2022)

Service and Outreach

2025–2026 **Organizer**, UO Student Geometry/Topology Seminar

Feb 2025 **Panelist**, UO Math Club event

Feb 2025 **Volunteer**, Eugene Math Festival
 Led educational booth on curvature for local K-12 students

2021–2025 **Mentor**, First-year graduate students, University of Oregon

- Sep 2023 **Co-Organizer**, “*Boot Camp*,” *University of Oregon*
Led 2-week mathematics orientation for incoming graduate students
- 2023 **President**, *UO chapter of the American Mathematical Society (AMS)*
Organized events for graduate and undergraduate students; managed chapter budget with the chapter’s treasurer
- 2022 **Diversity Chair**, *UO chapter of the AMS*
- 2021–2022 **Student Representative**, *UO Graduate Affairs Committee*
Attended committee meetings; organized events for incoming graduate students; handled office assignments and mentorship pairings
- 2020–2022 **Chair**, *Social and Professional Enrichment Committee, UO chapter of the Association for Women in Mathematics (AWM)*

Experience

- 2019–2026 **Graduate Student Employee**, *University of Oregon*
 - Instructor for fifteen university courses with consistently positive student evaluations; developed course materials and assessments; Anderson Teaching Award
- 2018 **Math Research Assistant**, *Seattle University*
Researched classification of domains in harmonic analysis and partial differential equations
- 2016–2018 **Consumer Product Advisor**, *Microsoft*
 - Helped customers find hardware and software solutions
 - Maintained detailed Microsoft Office knowledge
 - Provided professional technical support for personal and business consumers

Skills

- Scientific Computing in Mathematica, MATLAB, Python, and R
- Course Design
- Oral Presentation
- Academic Writing in LaTeX
- Program Leadership and Event Planning

Conferences and Workshops Attended

- Jun 2026 The Geometry and Physics of Higgs Bundles (Kashiwa, Japan)
- Jun 2026 New Developments in Kobayashi-Hitchin Correspondence and Higgs Bundles 2 (Osaka, Japan)
- Sep 2025 FRG Fall School on Quantization and Lagrangians (Chicago, IL)
- Jul 2025 Higgs Bundles and Harmonic Maps (Kiel, Germany)
- Jun 2024 SLMATH Summer School: Special Geometric Structures and Analysis (Moraga, CA)
- Jul 2022 Geometry and Physics of ALX Metrics in Gauge Theory (San Jose, CA)
- Jun 2022 Geometric Structures (re)United (Chicago, IL)

References

Available upon request